Rank-Adaptive Failure Summaries in Oracle Matroids

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Abstract

Fix a total order on an explicitly listed matroid of rank r, with at most one failure label per item. Let T be the sum of the numbers of new canonical-basis items introduced by each individual label failure. We construct one independent subset of the original canonical basis, of size $r - T$, whose contraction preserves every subsequent failure query after reinserting the surviving contracted items. The identity holds for every original order prefix, including rank drops. It reduces the canonical-basis tree from rank r to rank T, giving $O(N(r + T^h))$ independence queries for $T \geq 3$. We then give a direct rank-two algorithm with $O(N + h^2)$ queries and a matching worst-case lower bound for computing the exact canonical union when $N \geq h + 3$. At residual rank $T = 2$, the total query count becomes $O(Nr + (h + 1)^2)$, with the sharp retained size $r + 2h$. The rank-one residual requires only one input scan. An additional all-label-budget endpoint uses $N + r_U\ell$ queries at arbitrary rank. No linear representation of the original matroid is assumed. Oracle calls and ordinary computation have separate explicit bounds.

Part I

A fixed contraction for every failure query

Status: ready for review. The complete argument has undergone an independent internal proof review and exact finite checks. This is an oracle-matroid result for at most one label per item. It makes no representation assumption or literature-priority claim. The order is fixed throughout; a different cost order can require a different contraction.

1. Setup and the effective rank

Let M be a matroid on a finite explicitly listed ground set E of size N and rank r, with a fixed total item order. Each item has one label or is unlabelled. Distinct label classes D_λ are disjoint. A failed label set F leaves $E_F = E \setminus \bigcup_{\lambda \in F} D_\lambda$. Write B_F for the canonical greedy basis of this surviving ground, including its possible rank drop. Let B_0 be the unfailed basis, and set

$$b_0 = |\{e \in B_0 : e \text{ is unlabelled}\}|, \qquad b_\lambda = |B_0 \cap D_\lambda|, \qquad \delta_\lambda = r - r_M(E \setminus D_\lambda).$$

Every surviving item of B_0 remains accepted by greedy. Indeed, its surviving earlier prefix is a subset of its original earlier prefix and therefore cannot span an item that the original prefix did not span. Consequently

$$R_\lambda = B_{\{\lambda\}} \setminus B_0, \qquad |R_\lambda| = b_\lambda - \delta_\lambda.$$

In particular, labels absent from B_0 have $\delta_\lambda = 0$ and $R_\lambda = \varnothing$. Define

$$T = \sum_\lambda |R_\lambda| = r - b_0 - \sum_\lambda \delta_\lambda, \qquad 0 \le T \le r. \tag{1}$$

This is the single-failure replacement total, counting an item once for each single failure in whose basis it appears. The theorem below shows that it is also the rank of one fixed contraction sufficient for every failure budget.

Theorem 1 (fixed contraction). One can construct an independent set $Q \subseteq B_0$ of size $r - T$ such that, for the single fixed matroid $M' = M/Q$ on $E \setminus Q$, every failure set F satisfies

$$B_F = (Q \cap E_F) \,\dot\cup\, B'_F, \tag{2}$$

where B'_F is the canonical basis of $M'|((E \setminus Q)_F)$ in the inherited order. In particular M' has rank T. For every original order prefix P,

$$r_M(P \cap E_F) = |Q \cap P \cap E_F| + r_{M'}((P \setminus Q) \cap E_F). \tag{3}$$

The construction uses at most $(1 + \ell_0)N + r$ independence-oracle queries, where ℓ_0 is the number of labels meeting B_0, so $\ell_0 \le r$. It is deterministic. A query to the contracted independence oracle is one query to the original oracle on the queried set together with Q; ordinary query-set representation and input bookkeeping are charged separately.

2. Constructing the contracted set

Put every unlabelled item of B_0 into Q_0. For each label meeting B_0, compute a basis B_λ of $E \setminus D_\lambda$ in the fixed order. Extend this basis to a basis of E by scanning the items of $B_0 \cap D_\lambda$ in the fixed order, adding an item when independence permits it. This succeeds: B_λ spans $E \setminus D_\lambda$, and that set together with $B_0 \cap D_\lambda$ spans the original basis B_0 and hence all of E.

Let Q_λ be the added items. Then

$$Q_\lambda \subseteq B_0 \cap D_\lambda, \qquad |Q_\lambda| = \delta_\lambda, \qquad r_M((E \setminus D_\lambda) \cup Q_\lambda) = r_M(E \setminus D_\lambda) + |Q_\lambda|. \tag{4}$$

Thus Q_λ is independent over the entire complementary ground. Define

$$Q = Q_0 \,\dot\cup\, \bigcup_\lambda Q_\lambda.$$

All these items belong to B_0, so their union is independent. Equation (1) gives $|Q| = b_0 + \sum_\lambda \delta_\lambda = r - T$.

Computing B_0 and the ℓ_0 single-failure bases takes at most $(1 + \ell_0)N$ independence queries. The extension scans inspect at most $\sum_\lambda b_\lambda \le r$ additional items. No other label needs a query. The construction need not store all bases simultaneously: each B_λ can be discarded after its extension has been recorded. It stores the original input, B_0, Q, the actual basis-label list and one temporary basis.

3. Why failed contracted blocks are jointly independent

First, if a set I is independent over X, meaning $r_M(X \cup I) = r_M(X) + |I|$, it is independent over every $Y \subseteq X$. To prove this, order the items of I. Each has marginal rank one when added

to X and its predecessors. By diminishing returns for matroid rank, its marginal on the subset Y and those same predecessors is at least one, hence exactly one. Sum the increments.

Fix any failed label set F and write

$$Q_s = Q \cap E_F, \qquad Q_f = Q \setminus E_F = \bigcup_{\lambda \in F} Q_\lambda.$$

Add the blocks Q_λ for $\lambda \in F$ one at a time to E_F. When adding Q_λ, the ground already present consists of E_F and blocks Q_μ with $\mu \neq \lambda$. Every such item lies outside D_λ because label classes are disjoint. The present ground is therefore a subset of $E \setminus D_\lambda$. Equation (4) and the preceding paragraph show that this addition increases rank by exactly $|Q_\lambda|$. Thus

$$r_M(E_F \cup Q_f) = r_M(E_F) + |Q_f|. \tag{5}$$

This argument proves joint independence; separate independence of each block would not by itself be sufficient. It also proves the same equality starting from any subset of E_F, including a subset together with Q_s.

4. Contracting failed blocks commutes with their deletion here

For any $X \subseteq E_F \setminus Q$, equation (5) gives

$$\begin{aligned} r_{M/Q}(X) &= r_M(X \cup Q_s \cup Q_f) - |Q| \\ &= r_M(X \cup Q_s) + |Q_f| - |Q_s| - |Q_f| \\ &= r_{(M|E_F)/Q_s}(X). \end{aligned} \tag{6}$$

Hence the fixed contraction, restricted after failures, is exactly the same matroid as restricting after failures and contracting only the surviving members of Q. No general commutation of arbitrary overlapping contractions and deletions is being assumed. It is the complementary-ground independence in (4), and disjointness of the label classes, that justify (6).

By greedy persistence, $Q_s \subseteq B_F$. The canonical basis of a matroid is its lexicographically first basis in the fixed order. Bases of a contraction by an independent set Q_s correspond exactly to bases of the original matroid that contain Q_s, with those common items removed. Since B_F already contains Q_s and is first among all bases, $B_F \setminus Q_s$ is first among bases of the contraction. Removing common items preserves their lexicographic comparison. Together with (6), this proves (2).

A canonical greedy basis restricted to an order prefix is a basis of that prefix. Applying this fact to (2), in both the original and the contracted order, proves (3). This handles rank drops, unlabelled items, loops and repeated label classes without additional assumptions.

5. Exact failure-summary transfer and preprocessing cost

Let $H_h(M)$ be the union of all canonical bases under at most h failed labels. The usual prefix argument shows that it is the unique smallest original-item restriction preserving every surviving prefix rank: every item in the union has a witnessing prefix rank increase and therefore must be retained, while retaining all these bases suffices for every prefix. Taking unions in (2) gives the exact identity

$$H_h(M) = Q \mathbin{\dot\cup} H_h(M/Q), \qquad |H_h(M)| = r - T + |H_h(M/Q)|. \tag{7}$$

The unfailed query ensures that every item of Q occurs in the union. Labels that occur only in Q have no effect on the contracted ground. They may still change which fixed items are reinserted for a particular query; the original labels on Q are retained for this purpose.

Let $N' = N - |Q|$, and let ℓ' be the number of labels actually occurring in the remaining ground. Replace a binary-encoded budget h by $h' = \min\{h, \ell'\}$ for the minor construction. This is valid because failing additional absent labels does not change the minor. The algorithm therefore never loops a numeric h times solely because that integer has a long or huge binary value.

Apply the ordered-basis tree to the rank-T minor: compute its canonical basis at each node, output those items, and branch only on labels on that basis. There are at most T children and depth at most h'. Each node requires at most N' original independence queries through M/Q. Thus, for $T > 0$, a transparent total bound is

$$(1 + \ell_0)N + r + N' \sum_{j=0}^{h'} T^j. \tag{8}$$

It is $O(N(r + T^{h'}))$ queries for $T \geq 2$, and $O(N(r + h'))$ for $T = 1$. At $T = 0$, the minor consists only of loops and its tree is unnecessary: $H_h(M) = Q = B_0$ for every h. If the requested budget is zero, computing B_0 directly already takes at most N queries and no normalization is needed.

The tree's correctness is independent of the size-bound proofs. For a target failure set, either its remaining failures avoid the current canonical basis, in which case that basis remains canonical, or one of those failures is a branch label. Following such branches reaches a node whose basis equals the target basis. The fixed contraction only changes the rank controlling this branch count. A depth-first implementation stores the input, the fixed set Q, the output, and one rank-T basis/branch list per active depth. Each original oracle query can be represented explicitly using at most $r + 1$ items; this set-encoding work is not hidden in the count of oracle calls.

6. Rank-adaptive support bounds and sharp small-rank endpoints

Every existing ordered-prefix support bound at rank u can now be applied at $u = T$, then add $r - T$. For example, the proved Bernoulli prefix refinement gives, for $h \geq 2$ and one label per item (see the complete Bernoulli prefix refinement),

$$|H_h(M)| \leq r + \frac{h^h}{(h-1)^{h-1}} T. \tag{9}$$

The random product measure is only used in that earlier size proof; the construction here is deterministic. For one failure the sharp universal bound becomes $r + T$.

For two failures, the proved small-rank/profile theorem gives

$$|H_2(M)| \leq \begin{cases} r, & T = 0, \\ r + 2, & T = 1, \\ r - T + \min\{5T, \binom{T+1}{2} + 3\}, & T \geq 2. \end{cases} \tag{10}$$

In particular $T \leq 3$ gives $|H_2(M)| \leq r + 2T$, and $T = 4$ gives $r + 9$. These conclusions are insensitive to the original rank contributed by unlabelled or privately labelled directions.

There is a stronger all-budget endpoint at $T = 1$:

$$|H_h(M)| \le r + h. \tag{11}$$

A rank-one minor has at most one basis item at a node and hence at most one child. Its tree has at most $h' + 1$ nodes, so its union has at most $h' + 1$ items. Use (7). The bound is sharp: take $r - 1$ unlabelled coloops and a rank-one parallel class of $h + 1$ ordered items with distinct labels, for $h \ge 1$. The only one-failure replacement is the second parallel item replacing the first, so $T = 1$. Each of the first $h + 1$ parallel items is the first survivor when its predecessors' labels fail, so all $r + h$ original items are necessary. No representation is used by the algorithm or by the proof of the upper bound.

When the fixed order is nondecreasing cost followed by a permanent item ID, these summaries preserve the canonical minimum-cost basis under every allowed failure, including rank drops. This does not assert that the same Q is valid for all possible cost orders, or that a prefix-minimal summary is minimal among sets asked to preserve only the numerical optimum cost.

7. Exact independent finite checks

The checker constructs Q using only independence queries. Its reference uses ranks of the **original full prefixes** to identify canonical membership; it does not reconstruct the expected answer by the algorithm's contraction. For each failure it verifies joint independence of failed blocks, the rank identity (6) on every surviving subset for the small suite, equation (3) on every prefix, equation (2), and the complete union (7) for every failure budget. It also checks the stated query caps, $T = 0, 1$, rank drops, all-zero/loop instances, and the binary-large budget 10^{100}.

The completed small suite includes all 92 matroids on at most four labelled items, all one-label partitions up to label renaming with an optional unlabelled class, and every item order: 86,359 instances, 375,093 failure queries, 1,495,618 prefix comparisons and 1,450,577 arbitrary-subset contraction-rank comparisons. It checks 250,091 budget instances and 86,359 huge-budget cases.

The larger suite has 230 ordered/labelled instances from 220 matroids. These include binary represented matroids, the nonrepresentable 13-item obstruction augmented by privately labelled parallel components and unlabelled coloops, and exact $T = 1$ lower-bound families. It checks 4,680 failures, 42,512 prefixes, 196,560 additional contraction-rank comparisons and 1,051 budget instances. The larger arbitrary-subset checks use fixed-seed samples in addition to the empty and full surviving ground; only the small suite is exhaustive over all such subsets. The default command uses all matroids through three items plus the complete larger suite; `--full` also runs every four-item instance. The checker writes no repository files and rejects Python `-O`. These finite checks supplement the proof rather than replace it. Reproduction commands and the scope of the other suites appear in the verification note.

Part II

Optimal rank-two queries and residual-rank terminals

Status: complete proof with exact finite verification. This manuscript concerns one fixed item order and at most one failure label per item. Its oracle lower bound is for computing the

exact canonical union, equivalently the unique smallest restriction preserving every surviving prefix rank. It is not a lower bound for returning an arbitrary valid restriction. No literature-priority claim is made.

1. Statement and model

Let M be a matroid on an explicitly listed ground set E of N items, given by an independence oracle, with $\operatorname{rank}(M) \le 2$. Fix a total order on E. Each item has either one label $\lambda(e)$ or no label. For a set F of failed labels, let E_F be the surviving ground and let B_F be its canonical greedy basis in the fixed order. Rank drops are allowed. Given a nonnegative binary-encoded budget h, the required output is

$$H_h = \bigcup_{|F| \le h} B_F. \tag{1}$$

This is the unique smallest original-item restriction satisfying

$$\operatorname{rank}_M(H_h \cap P \cap E_F) = \operatorname{rank}_M(P \cap E_F) \quad \text{for every original-order prefix } P \text{ and } |F| \le h. \tag{2}$$

For completeness, if an item enters a canonical basis at its own position, its earlier surviving prefix does not span it; omitting that item would therefore violate (2) at that prefix. Conversely, the portion of a canonical basis in any prefix spans that prefix. These two facts prove the assertion.

Let ℓ be the number of distinct labels on nonloops, and put $k = \min\{h, \ell\}$. The algorithm discovers the nonloops before using k. Failures of labels occurring only on loops never matter.

Theorem 1 (rank-two construction). There is a deterministic algorithm that computes (1), together with one failed-label witness for each retained item, using at most

$$2N + \binom{k+1}{2} + (k+1)^2 \tag{3}$$

independence queries. Every query contains at most two items. After the fixed order and label indexing have been supplied, its ordinary work is $O(N(k+1) + (k+1)^2)$, and its storage is $O(N + (k+1)^2)$ item/label records. Label comparison and indexing costs are separate; deterministic sorting gives $O(N \log(N+2))$ comparisons if indexing is not supplied.

The output has the sharp size bound

$$|H_h| \le \min\{N, 2(k+1)\}. \tag{4}$$

If the rank is known to be at most one, N singleton queries suffice and the output has at most $k+1$ items. The rank-zero case returns the empty set without queries when that rank is supplied as a promise.

Theorem 2 (matching exact-summary query lower bound). For every pair of integers $h \ge 0$ and $N \ge h+3$, some rank-two, singleton-labelled instance forces every deterministic exact algorithm to use at least

$$\binom{h+2}{2} + N - h - 3 = \Omega(N + h^2) \tag{5}$$

independence queries. This holds even when its ordinary computation is free and queries of arbitrary cardinality are permitted. Together with Theorem 1, it gives worst-case query

complexity $\Theta(N + h^2)$ in this parameter range. The restriction $N \geq h + 3$ is part of the lower-bound statement; it is not silently extrapolated to budgets covering almost all input labels.

The algorithm extends the earlier ordered-basis tree at low residual rank. At rank two, its query bound replaces the tree bound $N \sum_{j=0}^{k} 2^j$ by $O(N + (k+1)^2)$. The polynomial term is additive to the input-size term. The theorem counts independence calls, not the running time of a hidden implementation of that oracle.

2. The possible first surviving items

Query every singleton to remove loops from consideration, retaining the ordered list of nonloops. Among these, scan in order and record the first item of each previously unseen label. Stop immediately after either the first unlabelled nonloop or the $(k+1)$st recorded labelled item. If neither occurs, continue to the end. Call the resulting ordered list A and write $a = |A| \leq k + 1$.

Lemma 3. Under every allowed failure set with at least one surviving nonloop, the first surviving nonloop belongs to A. Every item of A is the first survivor for some failure set of size at most k.

Proof. A labelled first survivor must be the first nonloop with its label: any earlier such item survives with it. Skipping $k + 1$ recorded distinct labels requires more than k effective failures. An unlabelled nonloop cannot be skipped. This proves the first claim, including when all recorded labelled items fail and no nonloop remains.

For an item $x \in A$, fail exactly the labels of preceding items of A. These are at most k distinct labels, none is the label of x, and no unlabelled item precedes x. Every earlier nonloop has one of those labels. Thus x is the first survivor. □

Let F_x denote this particular witness: the labels on the earlier entries of A. It is precisely the set of labels of all nonloops before x.

When the rank is at most one, the first surviving nonloop is the complete canonical basis. Thus $H_h = A$, proving the rank-one part of Theorem 1.

3. Capped lists outside the relevant parallel classes

Two nonloops are parallel when their pair is dependent. Parallelism is an equivalence relation: if x, y and y, z are dependent nonloop pairs, rank submodularity gives

$$\operatorname{rank}_M(\{x, y, z\}) \leq \operatorname{rank}_M(\{x, y\}) + \operatorname{rank}_M(\{y, z\}) - \operatorname{rank}_M(\{y\}) = 1.$$

Consequently x, z are parallel. A pair of nonloops from distinct classes is independent. In a matroid of rank at most two, the canonical basis consists of its first surviving nonloop and, if it exists, its first surviving item outside that nonloop's parallel class.

Classify the items of A into parallel classes by testing each new item against the already chosen class representatives. This uses at most $\binom{a}{2}$ independence queries. Let $\mathcal{C}$ be the resulting set of classes, let $c = |\mathcal{C}| \leq a$, and choose a representative $z_C \in A$ for each $C \in \mathcal{C}$.

For each class C, construct an ordered list L_C as follows. Among the nonloops outside C, record the first occurrence of each distinct label. Stop after the first unlabelled item outside C, or after recording $k + 1$ distinct labelled items, whichever comes first. Thus $|L_C| \leq k + 1$.

These lists can be built in one input scan without learning all the parallel classes. Keep a touched-label set and a closed flag for each C. For a nonloop e and an open list L_C:

1. If e is labelled and that label already occurs in L_C, do nothing.

2. Otherwise, if $e = z_C$, do nothing. In every other case query $\{z_C, e\}$.
3. If the pair is independent, append e to L_C. Close the list if e is unlabelled or if the list now has $k + 1$ entries.

Lemma 4 (query charging). The list scan uses at most $N + c(k+1)$ independence queries.

Proof. Every successful independence query appends one item to one list. There are at most $c(k + 1)$ such appends. A nonloop e can be parallel to at most one of the pairwise nonparallel representatives z_C. Consequently at most one of the queries involving this input item is dependent, regardless of which lists are already closed or which labels were skipped. The total number of dependent queries is therefore at most the number of nonloops, hence at most N. □

This charging is about oracle calls. The ordinary scan can inspect all $c \le k + 1$ list states for each input item, even when it makes no query. That is the $O(N(k + 1))$ term in Theorem 1. In particular the proof does not replace this ordinary work by $O(N + k^2)$.

4. Exact witnesses and extraction of the minimum summary

For $e \in L_C$, let $D_C(e)$ be the labels on entries of L_C strictly before e. No such earlier entry is unlabelled, because that would have closed the list. Since e occurs before truncation, $D_C(e)$ is exactly the set of labels on all earlier nonloops outside C.

For $x \in A \cap C$ and $e \in L_C$ later than x, define

$$W(x, e) = F_x \cup D_C(e). \tag{6}$$

Call the pair admissible if

$$|W(x, e)| \le k, \qquad \lambda(x) \notin W(x, e), \qquad \lambda(e) \notin W(x, e), \tag{7}$$

where a label-exclusion condition is vacuous for an unlabelled item. Return A together with every item e occurring in some admissible pair. Store F_x as the witness for $x \in A$, and $W(x, e)$ as the witness when an admissible pair first adds e.

Lemma 5 (exact extraction). This output is exactly H_h.

Proof. Every $x \in A$ has the witness from Lemma 3. For an admissible pair, failing $W(x, e)$ deletes every nonloop before x and every nonloop outside C before e, while preserving both x and e. Hence x is the first surviving nonloop and e is the first survivor outside its parallel class. They form the canonical basis, so $e \in H_h$.

Conversely, suppose e is the second item of a canonical basis for an allowed failure set F, and let x be its first item. Lemma 3 gives $x \in A$. Let C be the parallel class of x. Every earlier nonloop outside C must fail. In particular e is the first outside-C item with its own label, if it is labelled. At most k distinct outside labels can precede it, and no unlabelled outside item can precede it. Therefore $e \in L_C$. The same failure set contains F_x and $D_C(e)$, so $W(x, e) \subseteq F$; it preserves x, e. Conditions (7) follow. Every canonical basis has at most two items, which proves equality. □

For ordinary-time accounting, process one $x \in A$ at a time. Initialize an indexed set with F_x, scan L_C in order, test (7) using its current size and membership flags, then insert the next list label. Touched-label lists reset those flags in $O(k + 1)$ work. This is $O(k + 1)$ work per x, apart from copying the first witness for a newly retained item. There are at most $2(k + 1)$ retained items by Section 5, so explicitly storing all witnesses also takes $O((k + 1)^2)$ records and work.

There are N singleton queries, at most $\binom{a}{2}$ classification queries, and at most $N + c(k+1)$ list queries. Using $a, c \le k+1$ proves (3). Clipping h to ℓ ensures that a very large binary budget never causes a loop of length equal to that numeric budget.

5. The sharp retained size

The following is a size proof, not the algorithm used to construct H_h. It gives a short self-contained specialization of the represented prefix lift.

Represent each nonloop parallel class by a distinct rational number t_C and represent each item in that class by $v_e = (1, t_C)$. Loops have zero vectors. This represents every rank-at-most-two matroid, except that a one-class ground has rank one as it should. Choose distinct rational numbers z_λ for all labels and a distinct dummy z_0 for unlabelled items. Lift each column to

$$w_e = v_e \otimes (1, z_e, z_e^2, \dots, z_e^k) \in \mathbb{Q}^{2(k+1)}. \tag{8}$$

Take the canonical basis K of the lifted columns in the original order. It has at most $2(k+1)$ columns. For any effective failure set F of size at most k, the polynomial

$$p_F(z) = \prod_{\lambda \in F} (z - z_\lambda)$$

defines a linear map sending w_e to $p_F(z_e)v_e$. This image is zero exactly for failed nonloops, and is a nonzero scalar multiple of v_e for surviving nonloops. Every original lifted prefix is spanned by its columns in K, so applying this map shows that K preserves each surviving prefix span. By the necessity argument preceding Theorem 1, $H_h \subseteq K$. This proves (4), without constructing the representation or assuming access to it.

The bound is attained by two independent parallel classes, each containing $h+1$ ordered items with pairwise distinct labels, all labels distinct across both classes. Every item is needed: fail the at most h labels of its predecessors in its own class. The unfailed earlier items of the other class cannot span it. Thus all $2(h+1)$ items belong to H_h.

6. Independence-oracle lower bound

Let $p = N - h - 3$. Put p loop items first in the order, followed by core items $b_1, \dots, b_{h+2}, z$. Give every item its own distinct label. The base matroid M_0 is uniform of rank two on its $h+3$ core items, with those p loops adjoined.

The last item z is absent from $H_h(M_0)$: among its $h+2$ core predecessors at least two survive any h label failures, and these are independent. For every pair $\{b_i, b_j\}$ among those predecessors, define M_{ij} by making this pair one parallel class and leaving all other core items singleton parallel classes. It is again a rank-two matroid. Fail the other h predecessor labels. The pair survives with rank one and z becomes the second canonical item. Hence $z \in H_h(M_{ij})$.

The independence answers of M_0 and M_{ij} differ on exactly the two-element query $\{b_i, b_j\}$. All singleton core queries are true; all sets of at least three items are dependent; every query containing a loop is dependent unless it is empty. Therefore the transcript of a correct deterministic algorithm on M_0 must query every one of these $\binom{h+2}{2}$ pairs. Otherwise it has the same transcript on a matroid whose required output differs.

For each of the p initial loops u, also consider the variant in which u becomes a nonloop parallel to b_1, while every other item is as in M_0. The rank remains two. This item is the first nonloop in the unfailed order, so it belongs to the required summary in the variant and not in

M_0. A query distinguishing this variant from M_0 is either $\{u\}$ or a pair containing u and a core item not parallel to it. It can distinguish at most one such loop variant. Queries containing two of the original loops remain dependent in each single-loop variant, and queries of size at least three remain dependent by rank two.

The transcript on M_0 therefore requires at least p further queries, disjoint from the mandatory core-pair queries. This proves (5). It is an oracle-information lower bound, not merely an input-reading cost. The exact-minimum output requirement is essential: returning all of E always preserves prefixes without learning the oracle.

7. Fixed-contraction and residual-rank consequences

The fixed rank-loss contraction constructs $Q \subseteq B_0$ with $|Q| = r - T$ and the exact identity

$$H_h(M) = Q \,\dot\cup\, H_h(M/Q), \tag{9}$$

where T is the total number of single-label replacements counted with multiplicity, and the fixed minor has rank T. Its construction takes at most $(1 + \ell_0)N + r$ original independence queries, where ℓ_0 counts the labels meeting B_0. Write $N' = N - |Q|$, and clip the remaining failure budget to the labels on nonloops of the minor.

Corollary 6. Once that fixed contraction is computed, the exact summary uses no further queries if $T = 0$, at most N' further queries if $T = 1$, and at most

$$2N' + \binom{k+1}{2} + (k+1)^2 \tag{10}$$

further queries if $T = 2$. The corresponding support bounds are

$$|H_h(M)| \le r + Th \qquad (T \in \{0, 1, 2\}). \tag{11}$$

For $T = 1, 2$ and $h \ge 1$, equality is attained by adding $r - T$ unlabelled coloops to T independent parallel chains of $h + 1$ items with all labels distinct. Querying a singleton or pair in M/Q means querying its union with Q in M, so the number of calls is unchanged. The original query set may now contain $r - T + 2$ items; its encoding cost is not constant and is charged separately.

The same terminal algorithms may be used inside the ordered-basis tree, even when the original replacement rank is larger. At a node v with failed set F_v and remaining budget k_v, compute its canonical basis on the surviving ground. Stop immediately at budget zero. Otherwise, if its actual rank is at most two, call the corresponding algorithm above on that surviving restriction. Only larger-rank nodes branch on the labels of their canonical basis. The usual target-following proof remains valid: a target failure either leaves that basis untouched, or deletes a branch label; a terminal algorithm contains every target basis below its node.

Let $\mathcal{V}$ be the actual visited nodes of this truncated tree, N_v their surviving ground sizes, and $\mathcal{L}_2$ its positive-budget rank-two terminals. Including the rank-discovery scans, one explicit query bound is

$$3\sum_{v\in\mathcal{V}} N_v + \sum_{v\in\mathcal{L}_2}\left[\binom{k_v+1}{2} + (k_v+1)^2\right]. \tag{12}$$

The budget in each summand can be clipped again to that node's occurring nonloop labels. This is an actual-residual-rank bound; a tree whose surviving ranks remain large does not

receive an unproved polynomial guarantee. Rank-one and rank-two terminals replace whole failure subtrees by their exact canonical unions.

8. A linear-query endpoint when every label may fail

There is an additional endpoint when $h \geq \ell$, where now ℓ counts all occurring labels. It is valid at arbitrary matroid rank. Let U be the unlabelled ground and let $r_U = \mathrm{rank}_M(U)$.

Proposition 7 (all-label budget). If every subset of occurring labels is allowed to fail, the exact prefix summary can be constructed with at most

$$N + r_U \ell \tag{13}$$

independence queries. Thus it uses $O(N)$ queries at rank at most two. The query bound is an endpoint guarantee; it is not extended to smaller failure budgets without an additional argument.

Proof. Write D_λ for the items of label λ. A labelled item $e \in D_\lambda$ has some prefix witness if and only if it is accepted in the greedy scan restricted to $U \cup D_\lambda$. Indeed, failing every label except λ leaves the smallest possible earlier surviving prefix that still preserves e. Any witness on a larger surviving prefix continues to witness acceptance on this smaller prefix. The converse uses that very failure set. Similarly, an unlabelled item has a witness exactly when it is accepted in the greedy scan of U.

Maintain the current greedy basis B_U of the unlabelled prefix. For each label already encountered, also maintain the greedy basis B_λ of the prefix restricted to $U \cup D_\lambda$. When a label first occurs, initialize its state to the present B_U. A labelled input item is tested once against its label's state; retain it precisely when accepted. An unlabelled input item is first tested against B_U. If rejected, it lies in the span of the preceding unlabelled prefix and therefore in the span of every B_λ; no label state changes. If accepted, retain it and test its insertion into every active label state. Those independent tests update precisely the canonical greedy scans in the common original order. There are at most r_U accepted unlabelled items.

There is one principal test per input item and at most $r_U \ell$ extra state-update tests. Every returned item has the maximal-deletion witness identified above, and every possible prefix witness implies one of these greedy acceptances. This proves both exactness and (13). □

The stored bases use $O((\ell+1)(r+1))$ item and state records, in addition to input and output. An original independence query has at most $r+1$ items. With explicit query sets, ordinary basis and query-set copying has $O((r+1)(N + r_U\ell + \ell))$ work; this factor is separate from (13). For rank two these storage and ordinary-work bounds are linear in N. The endpoint explains why the quadratic lower bound is stated in the specific range of Theorem 2: a budget allowing every label to fail has a stronger direct algorithm.

9. Evidence and relationship to prior work

The finite checker implements the capped-list algorithm using only an independence callback. Its reference answer enumerates failures and computes membership from ranks of original prefixes, independently of the capped lists. It verifies every stored witness, the exact output, the query charging and the support bound. The full small suite covers every rank-at-most-two matroid through four items, every one-label partition with an optional unlabelled class, every item order and every effective budget. Additional larger partition matroids test repeated labels, long classes, late unlabelled items, rank drops, all-loop grounds, and very large binary budgets. A separate suite checks every oracle answer in the small lower-bound families. The all-label endpoint

is also compared with exhaustive failure unions on higher-rank represented matroids and the nonrepresentable 13-item example. These finite checks supplement Sections 2–8; they do not prove the asymptotic results.

Bentert, Fomin, Golovach and Morelle, *Fault-Tolerant Matroid Bases*, ESA 2025, Theorem 4 and Corollary 12, give an $O(N^2)$ algorithm for a different rank-at-most-two oracle optimization problem: select a minimum-weight set whose original rank survives any permitted number of individual-item deletions. The present output is the fixed-order canonical union under correlated label deletions and permits the target rank to drop. Their theorem is not used as a correctness or running-time premise here. This comparison identifies a nearby primary source, without claiming that a bounded search establishes priority for the present query bound.

Primary source: ESA 2025 article and bibliographic record.

Attribution and review scope. Yilin Liu and GPT-6 Astra Pro. *Algorithm Frontier*, 26 October 2026. The fixed order is part of the input contract; changing it may change the contraction. The independently computed finite rank tables supplement the written universal proof. No external peer review, proof-assistant formalization or literature-priority claim is made.